

Small Object Detection Using YOLO

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1. Introduction

This project investigates small object detection using the **YOLO (You Only Look Once)** model. The objective is to enhance the detection of small-sized traffic signs and lights from road images and video sequences, a task often challenged by **low resolution, occlusion, and high false positives**. To optimize detection accuracy, we performed four distinct training trials, varying key hyperparameters such as **epochs, image size, batch size, learning rate, augmentation techniques, and optimizers**.

The results from each trial were compared to assess the impact of different configurations on detection performance. This report documents the experimental setup, results, and conclusions drawn from the trials.

2. Dataset Description and Preprocessing

For training the YOLO model, multiple datasets of road signs were used, including:

- **US-Road-Signs-58** (used in Trials 1 & 2)
- **Traffic-Sign-Detection-YOLOv8-10** (Trial 3)
- **Mapillary Dataset** (Trial 4)

Preprocessing Steps:

1. **Image Resizing:** Images were resized according to the model's input requirement (e.g., 1024x1024, 1280x1280).
2. **Data Augmentation:** Techniques such as **flipping, rotation, scaling, and mosaic augmentation** were applied to improve generalization.
3. **Annotation Refinement:** Bounding boxes were validated to ensure accuracy in object detection.
4. **Normalization:** Pixel values were scaled to maintain consistency during training.

After an evaluation of dataset quality, the **YOLO-Final-Dataset (Trial 4)** showed the most promising annotation quality and variety.

3. Challenges in Small Object Detection

Detecting small objects like traffic signs presents several challenges:

- **Size Variation:** Small objects occupy very few pixels, making accurate detection difficult.
- **Occlusion and Clutter:** Traffic signs are often partially obscured by trees, vehicles, or other elements.
- **High False Positives:** Background elements were sometimes misclassified as traffic signs.

- **Computational Trade-offs:** Increasing image size improves detection but increases training time and memory usage.

4. Experimental Setup and Trials

We conducted four trials, each modifying key hyperparameters to optimize performance.

4.1 Trial 1

Objective: Baseline model using YOLOv8 with moderate augmentation and learning rate.

Key Hyperparameters:

- **Epochs:** 30
- **Image Size:** 1024
- **Batch Size:** 16
- **Optimizer:** AdamW
- **Learning Rate:** 0.005
- **Augmentation:**
 - Mosaic: 0.8
 - Flip Left-Right: 50%
 - Flip Up-Down: 40%
 - Scale: 0.4
 - Rotation: $\pm 5^\circ$

Findings: This model provided **moderate detection accuracy** but struggled with **very small objects** and false positives.

4.2 Trial 2

Objective: Increase training epochs and input image resolution to improve small object detection.

Key Hyperparameters:

- **Epochs:** 60
- **Image Size:** 1280
- **Batch Size:** 8
- **Optimizer:** AdamW
- **Learning Rate:** 0.001
- **Augmentation:**
 - Mosaic: 0.9
 - Scale: 0.5
 - Rotation: $\pm 10^\circ$

Findings: Improved small object detection, but **high memory usage and longer training time** due to increased image resolution.

4.3 Trial 3

Objective: Validate results from **Trial 1 with a different dataset** to test model robustness.

Key Hyperparameters:

- **Epochs:** 40
- **Image Size:** 1024
- **Batch Size:** 16
- **Optimizer:** AdamW
- **Learning Rate:** 0.005
- **Augmentation:** Same as Trial 1

Findings: Results were similar to **Trial 1**, reinforcing that moderate augmentation and hyperparameter tuning improve generalization.

4.4 Trial 4

Objective: Train with **higher epochs, larger batch size, and different optimizer (SGD)** to enhance small object detection.

Key Hyperparameters:

- **Epochs:** 100
- **Image Size:** 1024
- **Batch Size:** 32
- **Optimizer:** SGD
- **Momentum:** 0.937
- **Learning Rate:** 0.0005
- **Augmentation:**
 - Mosaic: Enabled
 - Scale: 0.5

Findings:

- **Best performing model** in detecting **tiny objects**, but **requires significant computational resources**.
- **Pretrained weights improved convergence speed** and reduced overfitting.

5. Results and Inference

Below are the detection results for each trial. **(Inference images will be inserted here).**

Trial	Epochs	Optimizer	Image Size	Augmentation	Small Object Detection Performance
1	30	AdamW	1024	Moderate	Fair
2	60	AdamW	1280	High	Good (High Memory Usage)

Trial	Epochs	Optimizer	Image Size	Augmentation	Small Object Detection Performance
3	40	AdamW	1024	Moderate	Similar to Trial 1 (But can detect small objects since dataset contained only small objects)
4	100	SGD	1024	High	Best Performance

Observations:

- Increasing **image resolution and batch size (Trial 2 & 4)** significantly improved detection accuracy but **required more resources**.
- Trial 4 performed the best** due to **higher epochs, batch size, and using SGD instead of AdamW**.
- Mosaic augmentation improved generalization** but required fine-tuning of other hyperparameters.

6. Inference Images

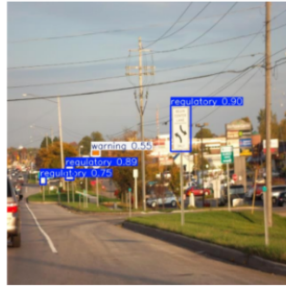
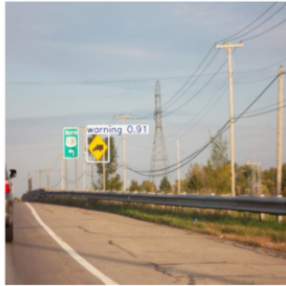
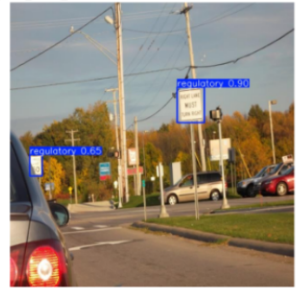
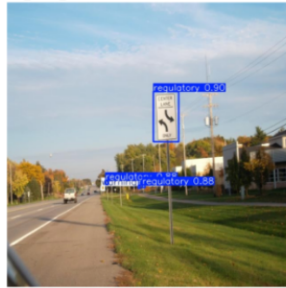
Trial 1 Result –



Trial 2 Result –



I_SAM_0886.JPG.rf.6c9e6ca7ec186bfd9d0a7ed50116394c0887.JPG.rf.ff10b1bb48143ef913834056181675AM_0888.JPG.rf.e339767dfcb7c948a0c3b91e0c08f5AM_0903.JPG.rf.8b21bb9a114e0728d25eaf0f40



I_SAM_0904.JPG.rf.1d0a39e1ed37c9c05326ed50116394c0887.JPG.rf.45391b091f1244b9e4fa656181675AM_0925.JPG.rf.f1eb1de2105b5fe352f4a53a27d54AM_0927.JPG.rf.b331c4d3732030a51f70e49c7e8



d_SAM_0955.JPG.rf.7acfcfac7296db951d008318e6d5116394c0887.JPG.rf.73628bf9c4eb2dae30a7b03e495b5AM_0969.JPG.rf.3d7d34fd8f5895a25df1a811e0c08f5AM_0979.JPG.rf.7ff2eb22a7d0e4a4c1db161355b



Trial 3 Result –



Trial 4 Result –

Click on trail 4 Video link below to see the results -

<https://drive.google.com/file/d/165KVw9jAVIEliifRTnoxAZFvMEMayve8/view>

7. Suggestions for Future Improvements

1. **Use Transformer-based Approaches – Applying Vision Transformers (ViTs) or Swin Transformers** could improve feature extraction.

2. **Optimize Anchor Boxes** – Customizing anchor sizes for small objects can enhance precision.
3. **Multi-Scale Feature Extraction** – Implementing **Feature Pyramid Networks (FPN)** may help retain details of small objects.
4. **Experiment with Different Backbones** – Using **EfficientNet or Swin-Transformer** as the feature extractor might boost performance.
5. **Post-Processing Adjustments** – Advanced **non-maximum suppression (NMS) techniques** can reduce false positives.

8. Conclusion

Through a series of **four trials**, we explored different hyperparameters and training strategies for **small object detection using YOLO**. **Trial 4 (epochs: 100, batch size: 32, SGD optimizer)** emerged as the best-performing model. The insights from this study highlight the importance of **image resolution, batch size, augmentation, and optimizer selection** in improving YOLO's ability to detect small traffic signs.

Future Work:

- Implement **real-time inference on embedded devices (Jetson Nano, Raspberry Pi)**.
- **Fine-tune anchor boxes** based on object size distributions.
- **Compare YOLO with Transformer-based models** for enhanced detection accuracy.